CDE4301: AI-Assisted Vibration-Based Condition Monitoring of Rotating Machinery for Predictive Maintenances



CDE4301 Final Report

Semester 1 AY2026/2027

Submitted by

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Acknowledgements

I would like to express my deepest gratitude towards my supervisor Prof Tang, for his constant encouragement and unwavering support. Despite his busy schedule, he goes the extra mile to accommodate consultations with me. His thoughtful insights and constructive feedback on the project has played a significant role in shaping the project into a product I had envisioned at the start of the FYP. His constant belief in my abilities gave me the confidence to complete the arduous year-long FYP. For that, I am truly grateful that Prof Tang is my FYP supervisor.

I would also like to thank the supervisors from the company I did my Industrial Attachment (IA) for providing me valuable opportunities to learn from their recent experiences, relating to vibration analysis. By addressing some of their enquiries and working collaboratively to solve certain real-world challenges, I have been able to investigate into areas beyond my research scope. This has truly broadened and enriched my research journey, giving me more insight to practical applications of vibration analysis.

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Table of Specifications

i) Frequency range:

- Sensor (ADXL356): Up to $\sim 1.5kHz$ per axis (X, Y, Z)
- Analog-to-Digital Converter (ADS1115): Up to $\sim 430Hz$
- Marine/plant Alternators: Mostly run at $25Hz$ or $60Hz$

ii) Acceleration range:

- Sensor (ADXL356): $\pm 10g$ or $\pm 20g$

iii) Noise floor/ Sensitivity:

- Sensor (ADXL356): $75\mu g/\sqrt{Hz}$

iv) Power requirements:

- Sensor (ADXL356): $2.25V - 3.6V$ (typically $3.3V$), $\sim 150\mu A$ per axis
- ADC (ADS1115): $2.0V - 5.5V$ (typically $3.3V$or $5V$), $\sim 150\mu A$ in continuous conversion mode
- Processing unit (Raspberry Pi 5): $5V$, $\sim 600mA - 5A$ (depending on load)

v) Form factor:

- Dimensions (Length×Breadth×Height): $102\ mm \times 82\ mm \times 40\ mm$

vi) Weight: $\leq 2kg$

Table of Abbreviations

i) RPi5: Raspberry Pi 5

ii) RMS: Root Mean Square

ii) FFT: Fast Fourier Transformation

iii) FIS: Fuzzy Inference System

iv) TSK: Takagi-Sugeno-Kang

v) IP: Ingress Protection

vi) ISO: International Organisation for Standardization

vii) FRF: Fundamental rotational frequency

viii) DE: Drive End

ix) NDE: Non-Drive End

x) TLS: Transport Layer Security

1. Introduction

1.1 Background on Industrial Vibration Monitoring

Industrial vibration monitoring plays a crucial role in many industries, from marine, oil and gas, manufacturing industries, to space and aircraft industries (Mohd Ghazali, Mohamad Hazwan, Rahiman, & Wan, 2021).

Reliability and efficiency of these machines are of utmost importance for continuous, uninterrupted running which minimises future maintenance costs. One of the most widely used modern techniques for condition monitoring is through vibration analysis, allowing for early detection of mechanical faults in rotating machinery such as, imbalance, misalignment and bearing failures.

The current vibration monitoring systems in the market are often expensive and requires specialised equipment and expertise. However, this limits accessibility especially for small medium enterprises. Through technological evolution, embedded systems and low-cost computing platforms such as Raspberry Pi 5 (RPi5), create an opportunity to develop affordable yet effective monitoring systems.

1.2 Problem Statement

Despite adopting vibration-based condition monitoring systems, there are many gaps in the current market solutions, such as accessibility and adaptability, particularly in marine and industrial rotating machinery systems (Chu, Nguyen, Yoo, & Wang, 2024).

In marine vessels and industrial plants, rotating machinery forms the backbone of power generation for essential critical operations. One of the most critical equipment onboard or in plants, are alternators. Alternators are the “heart” of a vessel or plant (Bright Hub Engineering, 2021). For marine alternators, supplying electrical power for navigation, propulsion support systems and onboard services. For plant alternators, supplying electrical power for plant’s internal operations depending on load requirements, ensuring stable energy output for machinery and process control systems, among other functions.

In a vessel/plant, there are multiple alternators so load can be placed on a different working alternator, should any breakdown. However, this avoids the issue, and the vessel/plant would have one less alternator, leading to a potentially dangerous scenario, where the other alternators might have issues too and breakdown. This results in downtime for the vessel/plant, and for vessel crew to be stranded at sea, as without alternators propulsion systems are disrupted.

Ultimately, alternators would have to be repaired, as using the remaining alternators while leaving one permanently down, is neither professional nor legal, due to class checks and audits for vessel/plants (Daya & Lazakis, 2024). However, cost of repair for alternators are expensive, due to extensive work required (E. Koh, personal communication, March 2026). Repair process of alternator found in Appendix under Appendix 1, A1.

While repair and transportation are costly, downtime costs vessels/plants the greatest (Daya & Lazakis, 2024; Jardine, Lin, & Banjevic, 2006). Unplanned downtimes cost Global 2000 $400 billion annually (Arda, 2026). In this period, employees' salaries are paid, while being unable to transport cargo (for vessels) or unable to produce goods (for plants). Such downtime disrupts supply chains and logistics, delaying delivery commitments and project deadlines. This may have a cascading effect where reputation is damaged and customers lose trust. For vessel downtime, it might affect port schedule, leading to legal implications. Vessel/plant management would want the alternator(s) to be repaired quickly, paying higher than market rate to have it repaired urgently.

With the critical issues highlighted above, come the market solution of vibration-based condition monitoring for predictive maintenance. Yet, despite availability of such systems, there are a few issues:

- High cost of commercial condition monitoring systems
- Limited adaptability to different machines and environments
- Lack of real-time, user-friendly visualisation tools
- Requires a trained analyst to determine issues from spectrum

Furthermore, many existing solutions rely purely on threshold-based methods, which have limitations in terms of robustness and accurate prediction of errors.

Threshold-based methods are effective for detection of severe faults under stable operating conditions (Cai et al., 2026). However, in real-world environments such as in vessel/plants, operating conditions may be dynamic due to variable loads and environmental influences. In such cases, the threshold-based approach may lead to false alarms or even missed early-stage faults. The Takagi-Sugeno-Kang (TSK) fuzzy inference system enhances robustness of the device in fault detection by modelling gradual transition between steady-state and transient states, allowing for more reliable detection under varying conditions (Haykin, 2009).

1.3 Value Proposition of AltGuard

AltGuard is proposed as a low-cost, real-time vibration monitoring system that addresses challenges through a hybrid approach.

The system integrates:

- Low-cost sensing setup using ADXL356 and ADS1115
- Real-time signal processing and feature extraction
- Hybrid fault detection algorithm combining threshold-based logic and TSK fuzzy inference
- Web-based dashboard for live monitoring and visualization

The novelty of AltGuard lies in its ability to:

- Target real-world rotating machinery applications (alternators in offshore marine/ onshore plants)
- Adapt to baseline machine behaviour using adaptive thresholds
- Differentiate between transient and steady-state conditions, improving reliability of fault detection
- Provide interpretable, real-time insights to users

2. Research Hypotheses and Objectives

2.1 Hypotheses

This FYP aims to research 2 different hypotheses:

1. Processed vibration data from sensors can reveal the “health” condition of machine.
2. Hybrid vibration-based condition monitoring system which integrates both threshold-based methods and TSK fuzzy inference model, can analyse “health” of machine better by achieving more reliable fault detection.

Thess hypotheses will be tested through both simulation-based experiments and actual scenarios in the NUS Central Workshop or at Megawatts Engineering, where I did my Industrial Attachment (IA) at.

2.2 Objectives

Overarching Objective:

To design and implement a real-time, ad-hoc vibration monitoring system capable of detecting common mechanical faults on-site, through analysis of vibration data, via implementation of a hybrid detection system that can detect faults during both transient and steady state of the machine. Example of faults are misalignment, imbalance and bearing faults. To visualise data better, development of a web-based dashboard showing real-time visuals of system outputs, such as graphs and final “health” classification, will be one of the objectives.

The specific aims of the AltGuard project are as follows.

Aim 1: Development of a real-time vibration data acquisition system

Aim 2: Implementation of signal processing and feature extraction algorithms

Aim 3: Developing a hybrid fault classification system

Aim 4: Implement transient and steady-state detection

Aim 5: Development of a real-time monitoring dashboard

Aim 6: Validate system performance with simulated data

Aim 7: Hardware integration for real-world deployment

Aim 8: Develop wireless data transmission capabilities

Aim 9: Improve system robustness and scalability

The detailed system specifications supporting these objectives are presented on Page 6 under “Table of specifications”.

3. System Design and Models

3.1 System Architecture Overview

The overall architecture of AltGuard can be visualised as the following pipeline:

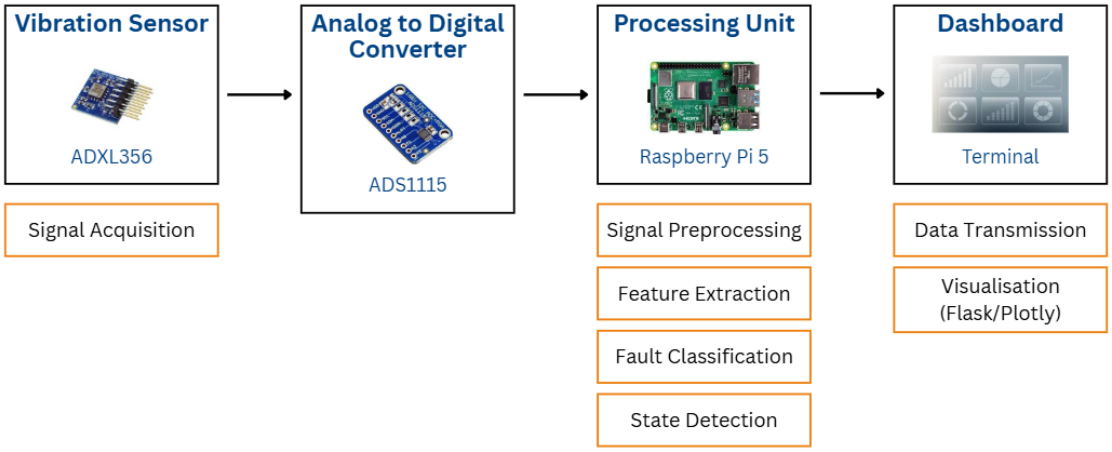


Figure 1: Overall system architecture

Vibration data is first captured through the accelerometer and converted into digital signals via an ADC. This data is then processed by the RPi5, where features are extracted and analysed. A hybrid classification model then determines the machine condition, and results are displayed on a web-based dashboard.

3.2 Vibration Signal Processing Model

The signal processing pipeline is designed to extract meaningful features from raw vibration data.

The key steps include:

1. Signal acquisition and preprocessing
2. Windowing and segmentation
3. Time-domain feature extraction
4. Feature-domain analysis using Fast Fourier Transformation (FFT)

The FFT is particularly important in identifying dominant frequencies and harmonic components, indicating specific mechanical faults.

3.3 Feature Extraction Techniques

Several features are extracted to characterise the vibration signal:

A) Root Mean Square (RMS): Indicates overall vibration energy. RMS is a baseline indicator that is widely used in vibration monitoring systems. It provides a simple but effective health metric of the overall system (Viking Analytics, 2024).
B) Kurtosis: Measures impulsiveness of signal, which can indicate sharp peaks in vibration, linking to bearing or gear faults (MathWorks, n.d.).
C) Dominant Frequency: Identifies primary vibration frequency for frequency spectrum analysis via FFT. This is standard in condition monitoring, with dominant frequency trend used for diagnosis (Viking Analytics, 2024).
D) Harmonic Ratio: Detects presence of harmonics for diagnosing misalignments and mechanical looseness. This is a key feature is predictive maintenance and signal processing systems (Towards Data Science, 2023).

These features are used as inputs for both threshold-based and fuzzy classification models.

3.4 Fault Classification Models (Threshold vs Fuzzy Logic)

Fault classification in vibration-based condition monitoring can be achieved using one of the following methods: Deterministic threshold-based method or Intelligent inference model.

Threshold-based classification

This is a widely used approach in industrial monitoring systems as it is both simple and reliable. In this method, predefined, hardcoded limits are set for key vibration features such as RMS, kurtosis and harmonic ratio.

For example:

- High RMS values may indicate imbalance
- Elevated kurtosis values suggest impulsive events such as bearing faults
- Increased harmonic ratios could indicate misalignment

The advantage in this classification method lies in its ease of interpretability, where there are set guidelines that these values should fall under (Jardine, Lin, & Banjevic, 2006). If they are above the thresholds, it is easy to pinpoint faults that lie within the system. However, it is limited by its lack of generalisability across different machines or operating systems. Furthermore, values that lies on the borderline of these thresholds, are not flagged by the system. This may cause potential sudden breakdown of monitored equipment due to rapid degradation. To improve on its weaknesses, we can use adaptive thresholds that are dynamically learned from baseline data through machine learning.

Fuzzy Logic Classification (TSK Model)

To overcome the limitations of rigid thresholds, a TSK fuzzy inference system is implemented (Zadeh, 1996).

Fuzzy logic allows for:

- Gradual transition between fault states
- Tackling uncertainty in identifying faults
- Greater flexibility in automated decision-making

In the TSK model, input features, such as RMS, kurtosis, harmonic ratio, are mapped to fuzzy membership functions (e.g. low, medium, high). These memberships are then combined through a rule base to generate a weighted output representing the machine condition.

For example:

- High RMS with low harmonic content → Imbalance
- High harmonic ratio → Misalignment
- High kurtosis → Bearing fault

Unlike Mamdani fuzzy systems, the TSK approach produces a mathematical output via weighted average, making it computationally efficient for real-time embedded systems (Zadeh, 1996).

Hybrid Classification Approach

The AltGuard system integrates both methods into a hybrid framework:

- During transient conditions, threshold-based classification is prioritised for rapid and reliable detection.
- During steady-state conditions, TSK fuzzy inference is used for more nuanced and adaptive classification.

This dual-layer approach improves overall system robustness by combining strengths of both deterministic threshold and intelligent inference methods.

3.5 Literature Review

Vibration-based condition monitoring has been extensively studied as a non-invasive method to detect faults in rotating machinery. It relies on the principle that mechanical defects produce characteristic changes in vibration signals. This can be extracted and analysed in both time and frequency domains.

Time-domain Feature Analysis

Specific time-domain features such as RMS and kurtosis are commonly used in fault detection (Viking Analytics, 2024). RMS represents the overall energy of the vibration signal and used in identifying imbalance. Kurtosis, measures impulsiveness of a signal and used to detect bearing faults due to its sensitivity to transient spikes.

Frequency-Domain Analysis

Frequency-domain analysis using FFT is a fundamental technique in vibration diagnosis. It allows the identification of dominant frequencies and harmonic components (Haykin, 2009).

For instance:

- Imbalance: Characterised by strong peak at fundamental rotational frequency (FRF)
- Misalignment: Strong peak at FRF and at additional higher harmonics of 2x, 3x
- Bearing defects: Broadband frequency components

Intelligent Fault Classification Methods

Fuzzy logic is a machine learning technique that is particularly useful, as it can handle uncertainty in measurements. Additionally, it is a robust and flexible way of providing interpretable rules which mimics engineering logic to guide the system in correctly classifying faults.

There are two main fuzzy inference system (FIS) models: Mamdani and TSK (Mamdani & Assilian, 1975; Takagi & Sugeno, 1985). While they are both fuzzy logic frameworks, they differ in handling outputs. TSK model is used in preference over Mamdani model as:

- Interpretability: TSK model is more mathematical and efficient, suited for analysis and adaptive models, while Mamdani is more intuitive and rule-based, which is close to human reasoning.
- Complexity: TSK model is computationally lighter and faster, however, Mamdani is computationally heavier due to defuzzification.

3.6 Research Gap

Despite advancements in vibration monitoring systems, several gaps remain:

- Many traditional systems rely heavily on threshold-based methods
- Fixed thresholds are often unable to adapt to varying machine conditions
- Requires manual in-depth analysis by trained specialists to identify faults via FFT spectrum
- Real-time, low-cost implementations using embedded systems are still limited

This project addresses such gaps by developing a hybrid system, integrating threshold-based rules and fuzzy inference methods, and adaptive learning, on a low-cost embedded platform, for automated reliable real-time alerts.

4. Hardware Architecture

4.1 Hardware Workflow Overview

The hardware architecture of AltGuard is illustrated below:

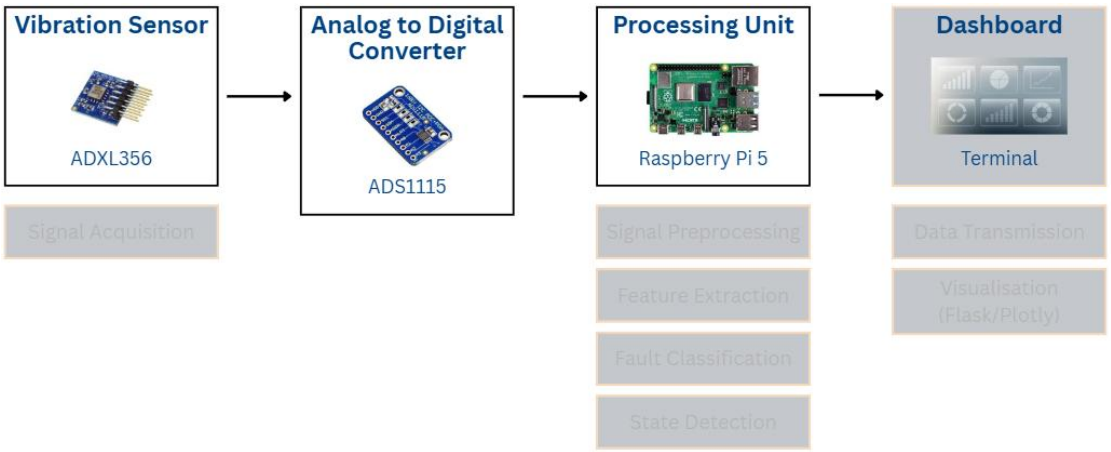


Figure 2: Overall hardware workflow

Analog vibration signals generated by the machine are first detected via the accelerometer. The analog output is then converted to digital signals using an ADC. These are then transmitted to the RPi5, where further signal processing and analysis are performed.

4.2 Selection of Components

Selection of components was based on cost due to limited budget. However, performance is another key priority, as vibration monitoring systems are crucial for a machine's life cycle and needs to accurately detect subtle faults early.

4.2.1 Accelerometer Selection

The ADXL356 was selected due to:

- High sensitivity suitable for vibration monitoring
- Ultra-low noise density of $75\mu g/\sqrt{Hz}$, enabling detection of subtle early faults
- Wide dynamic range, allowing detection of both small and large vibrations

Another accelerometer that was considered is the ADXL345. Although it is a digital accelerometer providing direct data output, and a lot cheaper than the ADXL356 (by roughly 65SGD), it was not selected as:

- Noise performance is moderate and less stable than ADXL356, with higher noise density of $220\mu g/\sqrt{Hz}$, making it suitable for regular consumer devices. However, for critical industrial usage, noise density should be as low as possible for most precise condition monitoring.

- ADXL345 has a lower range from $\pm 2g$ to $\pm 16g$ which are tuned for human-scale motion. However, ADXL356 has a range of $\pm 10g$ or $\pm 20g$, designed for industrial vibration and machinery condition monitoring, suiting AltGuard's objectives.
- Compared to digital accelerometers, analog output provides greater flexibility in signal analysis.

Furthermore, ADXL356 was chosen instead of other analog accelerometers such as MMA7361, as ADXL356 is a multi-axial accelerometer. Placing it at a global localised point or area would be an advantage since there is no need to change its location. Changing locations may give rise to variability in surfaces, thus affecting its accuracy.

Initial testing and research findings have included the ISO 10816-3:2022 standards, which are used to classify vibration severity in terms of velocity, mm/s RMS or in/s peak (ISO, 2022).

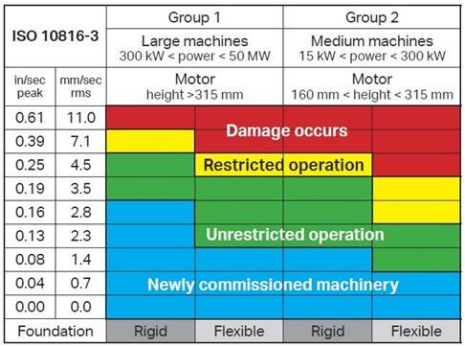


Figure 3: ISO 10816-3:2022 Standards

For large machines like alternators (>300 kW), the typical acceptable vibration levels are:

Good	$\leq 2.8 mm/s\ RMS$	$0.11g$
Satisfactory	$2.8 - 4.5\ mm/s\ RMS$	$0.17g$
Unsatisfactory	$4.5 - 7.1\ mm/s\ RMS$	$0.27g$
Unacceptable	$> 7.1 mm/s\ \ RMS$	$0.38g$

*The values in g are calculated assuming that the alternator operates at $60Hz$.

ADXL356 has a suitable range of $\pm 10g$ or $\pm 20g$ which is far above the expected vibration level ($< 1g$), ensuring no clipping even under severe fault conditions. With the ultra-low noise density ($75\mu g/\sqrt{Hz}$), it can detect subtle early faults that produces accelerations in the tens of milli-g range ($< 0.01g$).

Despite its compact size, its advanced capabilities render it industrially deployable.

4.2.2 ADC Selection

The ADS1115 was chosen for its:

- High resolution (16-bit), allowing precise measurement and hence, the detection of, small signal variations
- Compatibility with I2C communication, simplifying integration with the RPi5
- Cost-effectiveness compared to industrial-grade ADC systems

4.2.3 Processing Unit Selection

RPi5 was selected in comparison to the Pi 4 due to:

- CPU/GPU Performance: Pi 5 uses a quad-core ARM Cortex-A76 at 2.4 GHz while the Pi 4's Cortex-A72 at 1.6 GHz. This means that the Pi5 is roughly two to three times faster in benchmarks. High-frequency data can be acquired by the Pi 5 faster and it can deal with higher sampling rates without bottlenecks.
- Memory Bandwidth: Pi 5 has LLPDDR4X RAM with higher throughput, improving multitasking and data-heavy workloads.
- Storage Speed: Pi 5 supports PCIe for NVMe SSDs, offering much faster read and write speeds as compared to Pi 4's microSD-only setup
- Thermal management: Pi 5 comes with active cooling support and a power button, which helps to support continuous operation, however, Pi 4 runs hot under load. Thus, Pi 5 is better suited for 24/7 monitoring.

Despite Pi 5 costing around two times the price and consuming more power than Pi 4, the benefits that Pi 5 provides, greatly outweighs that of Pi 4, when considering the context of high-frequency vibration-based condition monitoring, that relies heavily on processing real-time data.

4.2.4 Connector Considerations

Molex connectors are considered over Dupont connectors as:

- Molex connectors are more reliable and durable.
- Molex connectors produce less noise, due to sturdier connection

These connectors may be a small component of the system, however, it affects accuracy and precision. Choosing a cheaper but noisier connector such as Dupont connectors, may lead to future errors when testing actual machinery.

4.2.5 Display and User Interface

A 2.4-inch SPI TFT display was selected to display local output by the system.

A screen can present numerous benefits over a traffic-light LED arrangement. While both are able to show operating status, the screen show graphical results and presents numerical vibration data received by the sensing unit. Thus, when remote dashboard is unavailable, on-site engineers can still look at the data and graphs from the screen. The screen can also show the potential underlying issue and confidence levels for easy analysis.

While a traffic-light LED arrangement is less costly and uses less power, the benefits of a screen outweighs those of the LED signals, as battery can be charged and the price difference is rather insignificant.

4.2.6 Power supply

The prototype's power architecture uses a 3.7V lithium-polymer battery with a Mini560 Pro buck-converter module to provide a regulated supply rail of 3V, which is the electrical requirements for all other modules in the system.

4.3 Enclosure and Mechanical designs

The enclosure is intended to suit marine offshore vessels/ onshore industrial plants, the device must be encased appropriately to match the environments.

Through thorough research, AltGuard's case should be IP67 (IEC 60529). IP67 is dust-tight, preventing not only dust, but salt crystals in marine environment or fine particles in plants, from entering the device with the sensor and ADC. Contamination may lead to potential noisy artefacts that degrade the system's accuracy.

Potential scenarios where water enters the plant/vessel, resulting in device immersion, IP67 rating prevents the interior, sensor and ADC, from coming in contact with water, thus preventing damage.

As saltwater is highly corrosive, IP67 sealed housing largely prevents ingress and prevents long-term degradation due to its corrosion-resistant material.

However, an IP67-rated sealed housing was not used in the FYP. This is due to the sensitive nature of the sensor. It should be as close as possible to the surface of the alternator to obtain the most accurate output. Commercially sold IP67-rated housings in the market are bulky and the walls are thicker for outdoor usage, which meant that the sensor would likely be unable to pick up accurate vibration readings from the alternator.

To ensure the enclosure is as waterproof as possible, I have incorporated an O-ring groove into the enclosure design to accommodate a seal between the mating surfaces. When the enclosure is closed, the compressed O-ring reduces the ingress of moisture and dust.

Furthermore, as the switch is found on the outside of the enclosure, an M12 silicone gasket is used to ensure no or minimal water or dust enters the device.

An honest disclaimer – the effectiveness of the arrangement would require extensive ingress testing before classifying is as waterproof or assigned an IP67 rating.

5. Software Architecture

5.1 Software Workflow Overview

The software architecture is designed in the following pipeline:

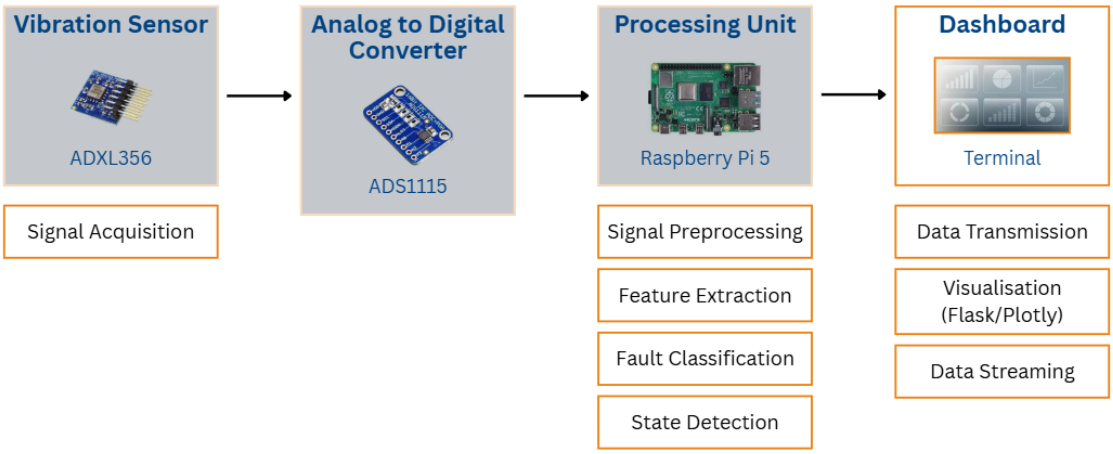


Figure 4: Software workflow overview

Each stage is implemented as a separate module to improve scalability and can be fine-tuned individually.

5.2 On-Machine Processing (Edge Computing)

On-machine processing includes:

- Signal preprocessing (DC removal, windowing)
- Feature extraction (RMS, kurtosis, frequency analysis)
- Fault classification using threshold-based and fuzzy logic methods
- Transient and Steady-state detection

More information on the above on-machine processing, such as codes and validation of methods used are found in the appendix under A2.

5.3 Off-Machine Communication and System Integration

Off-machine processes involves communication between the embedded system and external devices.

This would include:

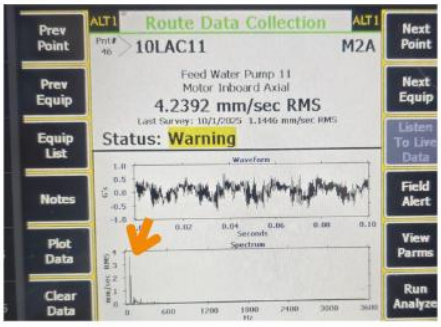
- Transmission of processed data to web-based dashboard
- Visualisation using Flask and Plotly
- Potential future integration with remote monitoring system

More information on the above off-machine processes can be found in the appendix under A3.

6. Initial Testing and Prototyping

6.1 Case Study: Megawatts Engineering

During the initial phase, after reaching out to Megawatts Engineering, they sent this image of a customers' vibration analyser.



2800 RPM Water Pump

Figure 5: Displayed results from analyser

Upon researching on vibration data analysis techniques, I came across some graphs, which helped in understanding the use of spectrum analysis.

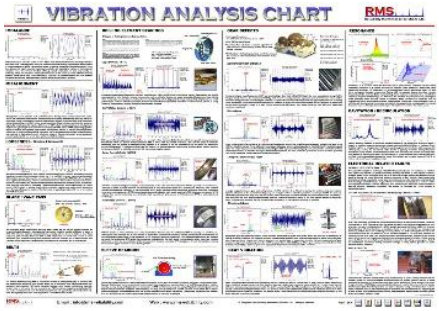


Figure 6: Vibration Analysis Chart

Through this, I was able to identify the associated fault with the system and propose it as a possibility. This turned out to be true eventually, as it was an "imbalance" issue and a part of the rotor had to be removed and balanced before reinstallation.

6.2 Simulation Results

Simulation results can be found in the appendix under A4.

7. Iterative Development and Validation

Through the course of the FYP, there have been countless of hardware and software design iterations. However, only crucial iterative milestones have been selected and be elaborated on.

7.1 Hardware and Enclosure Design Iterations

The hardware of the device is not only the face of the device, but also to protect the interior, where it houses the electronics necessary for vibration analysis. The following subsections elaborate on the hardware iterations leading towards the final product with regards to: Initial design issue, design modification and, engineering rationale and considerations.

7.1.1 Enclosure Size and Removal of Router

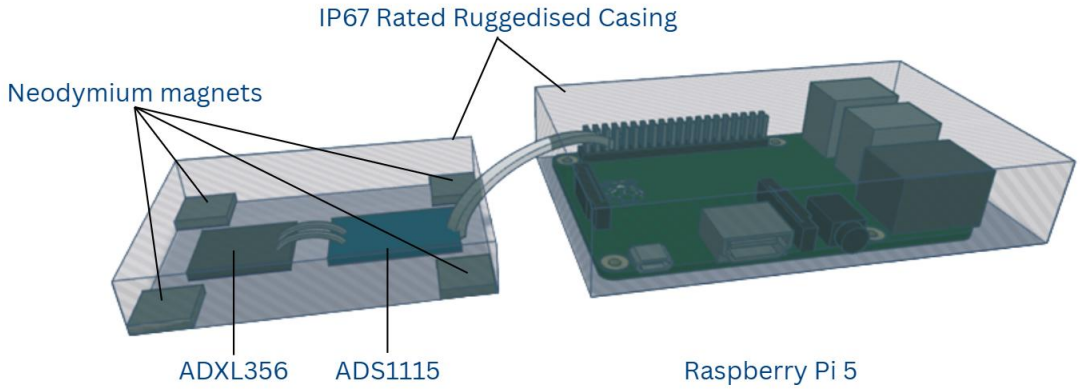


Figure 7: Before removal of router

Initial Design Issue: The initial concept included a dedicated wireless router. However, it is not necessary as the ESP32-S3 already supports Wi-Fi connectivity and the Wi-Fi routers are usually located in the control room, within the engine room, where the alternators are located (Megawatts supervisor, interview, 2026). Furthermore, a mini router would also take up a substantial amount of space within the enclosure.

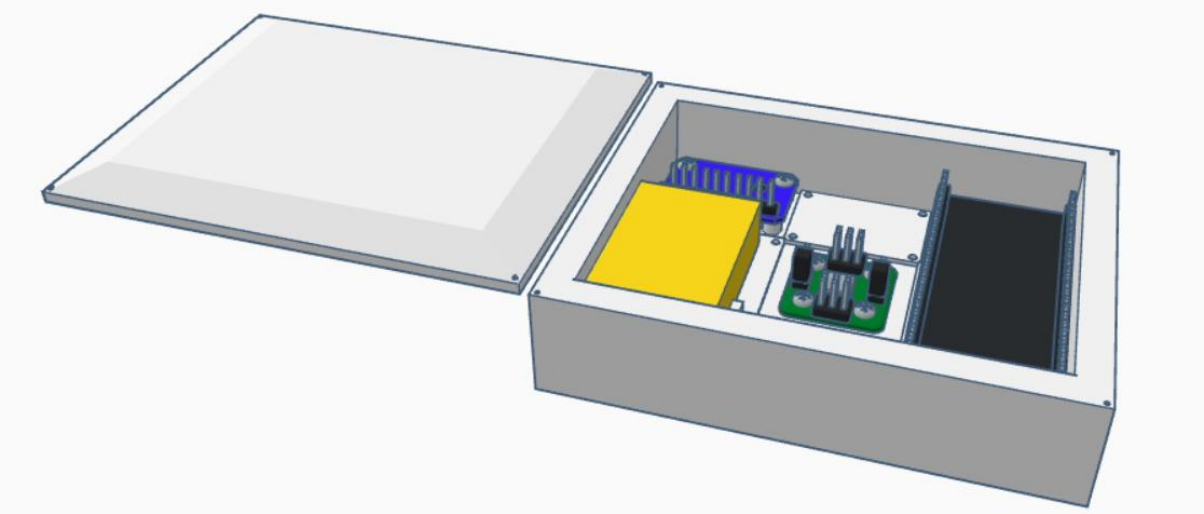


Figure 8: Removal of router

Design Modification: The router was removed from the system which enabled the enclosure to become more compact.

Engineering Rationale and Considerations: This change reduces total volume and avoids future unnecessary networking hardware which may interfere with existing Wi-Fi infrastructure onboard. For vessels without Wi-Fi or insufficient coverage in the engine room, we can always provide add-on services involving setting up Wi-Fi or routers for the vessel.

7.1.2 Enclosure Sealing and Environmental Protection

Initial Design Issue: After reducing the size of enclosure, as seen in Figure 21, the revised design did not include a dedicated sealing mechanism. This presented a potential weakness for the enclosure, as dust and moisture may be able to enter its interior.

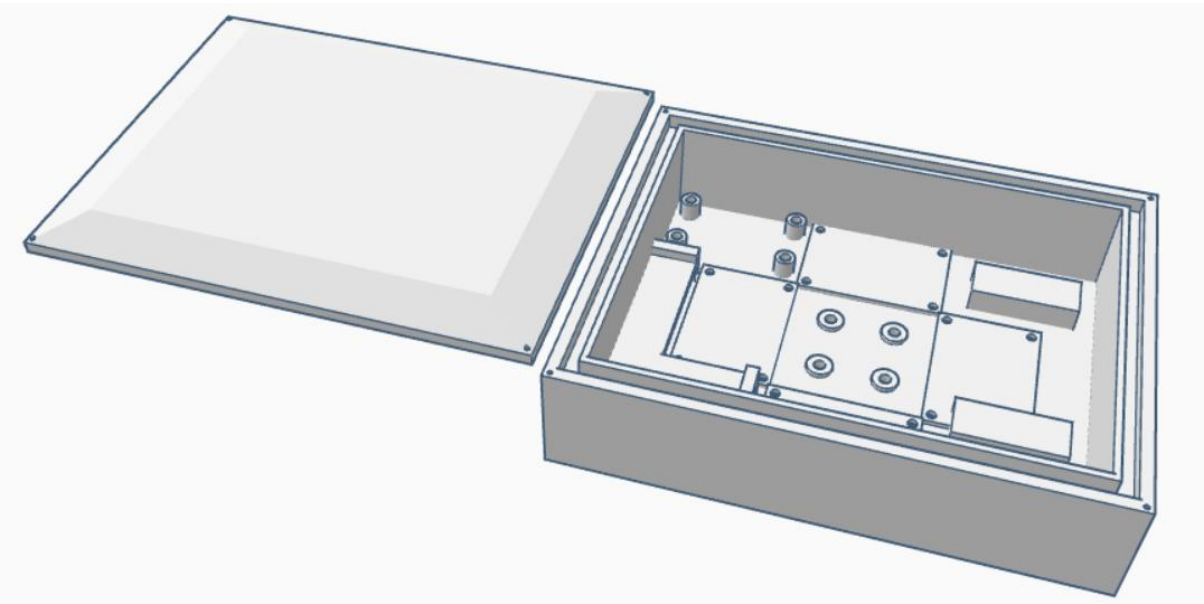


Figure 9: O-ring groove incorporated

Design Modification: An O-ring groove was incorporated into the enclosure design to accommodate a sealing ring between the 2 mating surfaces, the lid and the enclosure.

Engineering Rationale and Considerations: As the O-ring is compressed, it reduces the probability of water and dust entering its interior. However, the sealing design has not been demonstrated to be waterproof as it would have to be sent for ingress testing first. It would depend on a couple of aspects such as, but not limited to, groove dimensions, O-ring material and compression, and fastening arrangements. To ensure it is water-resistant or IP-rated, an appropriate ingress test has to be conducted before any claims are made.

7.1.3 Addition of Local Status Display and Shape of O-ring groove

Initial Design Issue: The original design lacked any indication of the device's operative status. Without local feedback, it will be difficult to determine whether the monitoring system has been powered properly or has connected to the Wi-Fi successfully. Furthermore, the shape of the O-ring groove made it difficult to insert an O-ring.

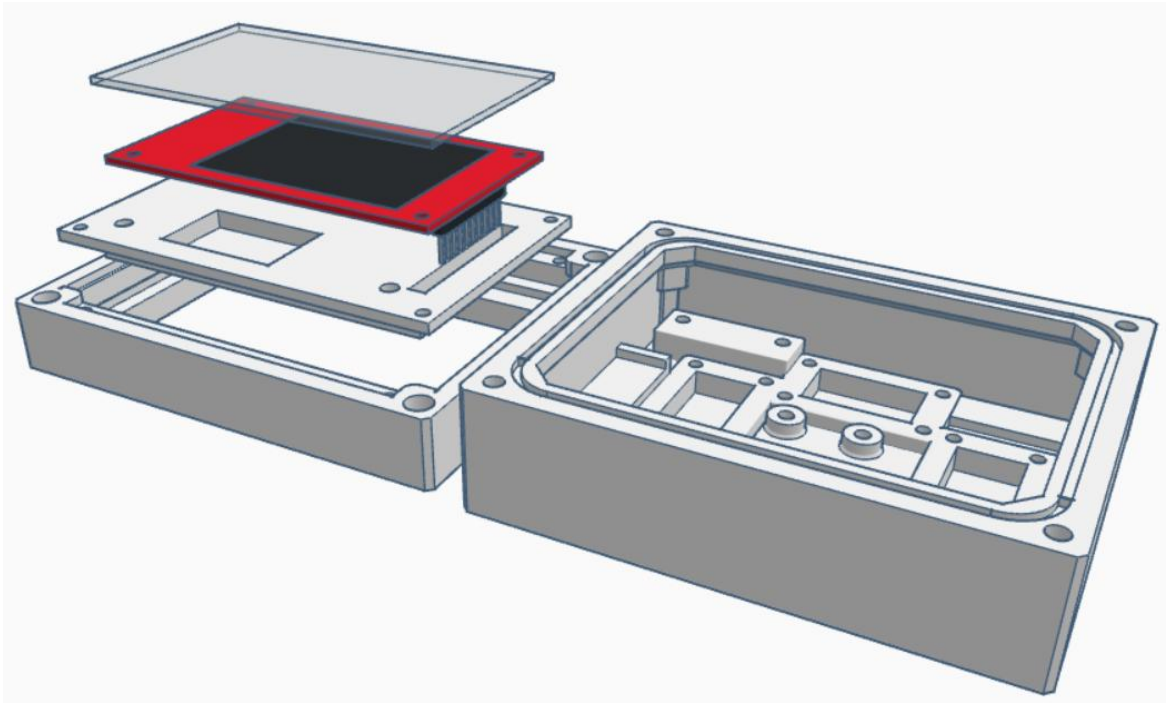


Figure 10: TFT Display and change in O-ring groove design

Design Modification: A small TFT display has been incorporated into the system instead of conventional LED signals. It is able to provide more informative and visual feedback, from startup status to sensor acquisition status to analysis of the issue. The O-ring groove was designed to be more circular to ensure the O-ring fits appropriately.

Engineering Rationale and Considerations: A display would make troubleshooting easier, as the error would be shown on the screen. Additionally, it is able to communicate numerical measurements and diagnostic information rather than just 3 different status categories without information on specific systemic flaws.

7.1.4 Addition of Physical Power Switch

Initial Design Issue: In the original design, connecting of the battery would power the circuit up. However, switching it off would require opening the lid and disconnecting the battery from the circuit. This brought much inconvenience during testing phase.

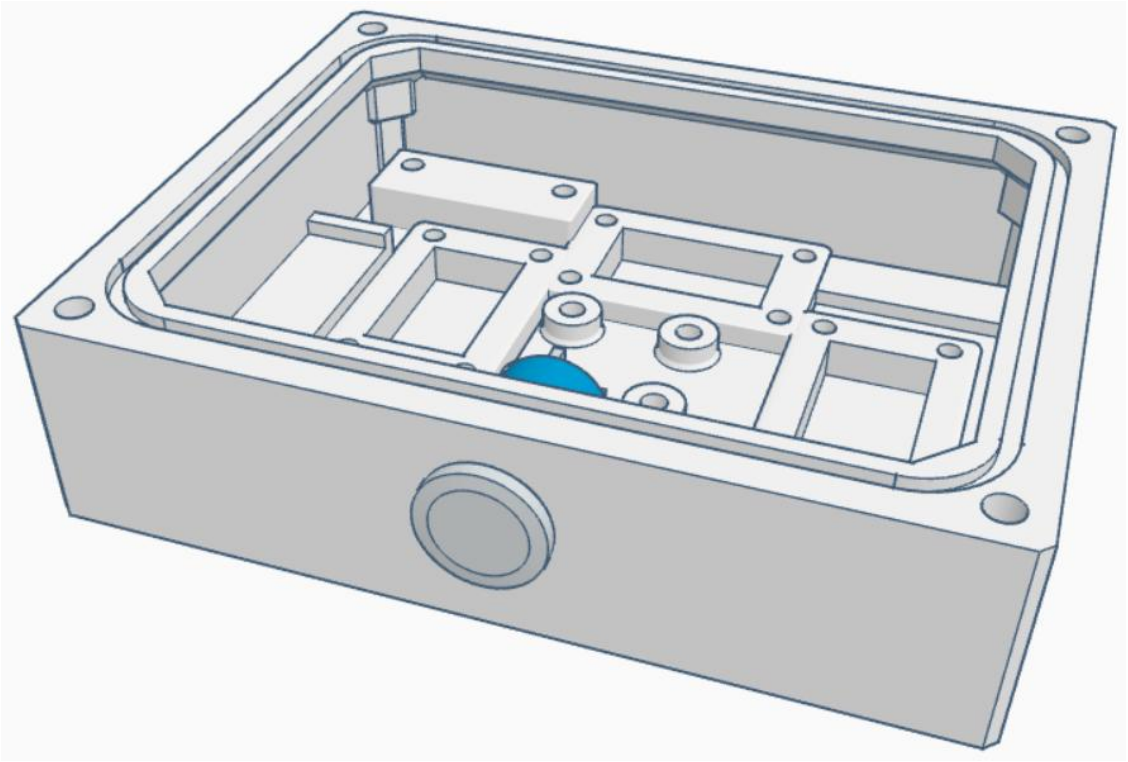


Figure 11: Addition of Power Switch

Design Modification: The enclosure was widened to accommodate a latching power switch that can be easily accessed externally.

Engineering Rationale and Considerations: This improved convenience and reduces the need to handle essential wiring near the battery, reducing the risk of electronic malfunction from mishandling critical wiring. Taking the need to mitigate dust and water ingress into consideration, a slot was made under the switch to hold a silicone gasket between the mating surfaces of the switch and enclosure.

7.2 Software Development and Debugging Iterations

Developing software from scratch entails multiple trial-and-error, thus debugging and improving the codes are critical for a functioning AI-assisted vibration analyser. The following subsections elaborate on the key changes made to the software through: Problem identified, corrective action and significance.

7.2.1 Correction of I²C pin configuration

Problem Identified: During the ESP32-S3 and ADS1115 integration, the configured I²C pins did not match the physical wiring. The intended connections were SDA and SCL on the ADS1115 to GPIO9 and GPIO8 on ESP32-S3, but an earlier code used the opposite order in `Wire.begin()`

Corrective Action: The I²C initialisation was corrected to `Wire.begin(9,8)` which matches the correct connections. To check if ADS1115 was initialised successfully, startup diagnostics were programmed in the ESP32-S3.

Significance: Consistent link between hardware assignment and software configurations is crucial. Correcting the pin configuration helps address and avoid a potential communication problem.

7.2.2 Introduction of Simulation for Software Development

Problem Identified: While integrating the sensor hardware, it is difficult to debug and to create particular fault-like signals.

Corrective Action: A simulation interface (`adc_interface.py`) was built and incorporated into the system to generate common faults, such, normal, imbalance, misalignment and bearing-fault signals. Simultaneously, the conditions are being processed by `main.py`.

Significance: Simulation makes it easier to reproduce known signal characteristics. For instance, to simulate misalignment conditions, we can introduce a second harmonic. However, while simulations are helpful in debugging, they are simplified representations of fault-related vibration signals, rather than experimentally validated, hence further testing is required to prove realistic diagnostic accuracy on actual alternators.

7.2.3 Development of Modular Signal-Processing Pipeline

Problem Identified: The system requires more raw acceleration readings from the ADXL356BZ to distinguish between the distinct potential machine conditions. To convert these acquired vibration windows into meaningful features, we would have to use classification algorithms.

Corrective Action: The processing pipeline was then organised into unique modules. The processing.py module performs signal processing, feature extraction and classification. The `fuzzy.py` module handles TSK inference logic and lastly, `utils.py` provides adaptive threshold utilities.

Significance: Separating acquisition, feature extraction, classification and utility functions help make coding easier to test and debug, before integrating everything into a complete monitoring system.

7.2.4 Correction of Communication and Integration Issues

Problem Identified: The communication configuration between ESP32-S3 and RPi5 needed further debugging. Initially, the ESP32-S3 was at `192.168.198.249`, while the RPi5 was at `192.168.121.57`, belonging to different IP subnets, hence, preventing communication. Furthermore, the Flask dashboard used port 5000 to receive data, but the ESP32-S3 uses TCP port 5001 to send data.

Corrective Action: Checking and editing the subnets and port numbers in the code.

Significance: Network configuration is crucial in establishing communication and should be checked against devices' current IP addresses and subnet masks. An initial integration test needs to be conducted to safely establish the communication between the ESP and RPi5. This entails that RPi5 can received transmitted data, parses and process the data correctly and ultimately, updating the dashboard with the values and spectrum.

7.2.5 Dashboard debugging and FFT Visualisation

Problem Identified: The dashboard required debugging during development. One of crucial errors involved a duplicate Flask endpoint for `/api/fft`, causing an `AssertionError`, as the endpoint name was repeated. Another critical issue was the frequency-domain visualisation which merely displayed a lot of frequency bins, rather than meaningful spectral peaks.

Corrective Action: The duplicate endpoint was resolved by retaining only one route definition. Whereas, the FFT display was further developed to use both frequency and magnitude arrays that were generated by signal-processing module. Thereafter, peak detection is used to display prominent spectral components as markers on the frequency spectrum.

Significance: Resolving duplicate route definitions removes the endpoint conflicts, which results in initialising the Flask application successfully. Peak markers can make detection of frequency peaks easier to interpret during fault investigation.

7.3 Important Software Limitations for Discussion

7.3.1 Sampling-rate Limitation

The ADS1115 has a maximum data rate of 860 samples per second. However, the ESP32 does not guarantee 1000 accurate samples per second, especially since ADC conversion takes time. Hence, limiting usable frequency range and constrain high-frequency bearing diagnostics. This also results in delays in updating dashboard. Although a faster ADC is recommended to measure more samples per second, due to the limited FYP budget, the ADS1115 was the most affordable and had the best performance for its price point.

8. Testing and Results

8.1 Testing Methodology and Setup

The aim of testing AltGuard is to determine its ability to measure mechanical vibration and examine its ability to identify faults accurately. The testing was done in two distinct phases.

The first phase involved physical vibration measurements against a commercial vibration meter and at different locations. Measurement were conducted by both a commercial vibration meter and AltGuard, at the drive end (DE), non-drive end (NDE) and an additional accessible flat surface of the motor housing.

The second phase involved frequency-domain analysis using extracted data from the first-hand testing and with extracted data from Zenodo, which provided datasets of different faults in a generic rotating machinery (Marra & Garcia, 2024). This allows the evaluation of FFT processing pipeline and TSK membership functions to be evaluated without the need to introduce faults into the motor.

8.2 Comparison of Vibration Measurements at Different Locations

8.2.1 Vibration Results

The commercial meter used to provide the following readings were provided by Megawatts Engineering. It was a METRAVI portable vibration meter.

Three measurement locations were selected:

Location 1: DE

Location 2: NDE

Location 3: Any available flat surface

Table 1: Vertical vibration measurements

Measurement location	Location 1	Location 2	Location 3
Commercial meter (mm/s)	0.8	0.8	-
AltGuard meter (mm/s)	0.79	0.81	0.77
Percentage difference (%)	-1.25%	+1.25%	-

Table 2: Horizontal vibration measurements

Measurement location	Location 1	Location 2	Location 3
Commercial meter (mm/s)	1.5	1.4	-
AltGuard meter (mm/s)	1.49	1.42	1.45
Percentage difference (%)	-0.67%	+1.43%	-

Table 3: Axial vibration measurements

Measurement location	Location 1	Location 2	Location 3
Commercial meter (mm/s)	0.8	0.8	-
AltGuard meter (mm/s)	0.81	0.8	0.78
Percentage difference (%)	+1.25%	0%	-

8.2.2 Analysis and Evaluation

These results show slight deviations in vibration measurement by AltGuard and the commercial vibration meter. It also shows that AltGuard is able to produce relatively close readings, with the absolute difference being 0.00mm – 0.02mm/s.

The small discrepancies to the vibration picked up by the sensor could be due to sensor placement, mounting contact, orientation of sensors and the transmission of mechanical vibration through the motor body. The sensor in the commercial vibration meter uses a single-axis accelerometer, which means that it had to be removed and repositioned in axial, horizontal and vertical positions for measurement. This could give rise to slight differences in vibration due to mounting contact and orientation of sensors.

As the commercial meter was not placed in the same third location, there is no numerical benchmark. However, placing AltGuard on any flat available surface, showed a lower amplitude in vibration results, possibly due to its distant relative position to the bearings and, the DE and NDE.

8.3 Benchmarking and Reliability Evaluation

The benchmark equipment used for reference is the above commercial vibration meter.

The results provide some evidence that the vibration velocity output from AltGuard has relatively minute differences from that of the commercial meter, supporting its data acquisition capabilities.

However, the benchmark is limited as this commercial vibration meter was unable to provide a frequency-spectrum output. Thus, there was no graphical benchmark for comparison, regarding the frequency-spectrum output.

Distinguishing the two aspects for performance:

1. Accuracy of vibration acquisition: There was close agreement between AltGuard and the commercial vibration meter.
2. Condition-monitoring reliability: Unable to establish reliability in predicting faults, since the motor used for testing was post-repaired, hence was in a healthy state. It was not earlier verified before by a commercial vibration meter with spectrum output to exhibit targeted fault conditions.

8.4 Graphical Comparison of Healthy and Faulty Vibration Results

To evaluate fault-detection capabilities of AltGuard, the frequency-spectrum outputs by the healthy primary data and extracted secondary data can be analysed by the AltGuard's Flask-based dashboard. There are two objectives:

- Determine if TSK inference model can distinguish between healthy and faulty motor conditions
- Assess if the FFT graph can pick up peaks that are indicative of the correctly associated fault

8.4.1 Frequency-spectrum output Analysis from healthy motor

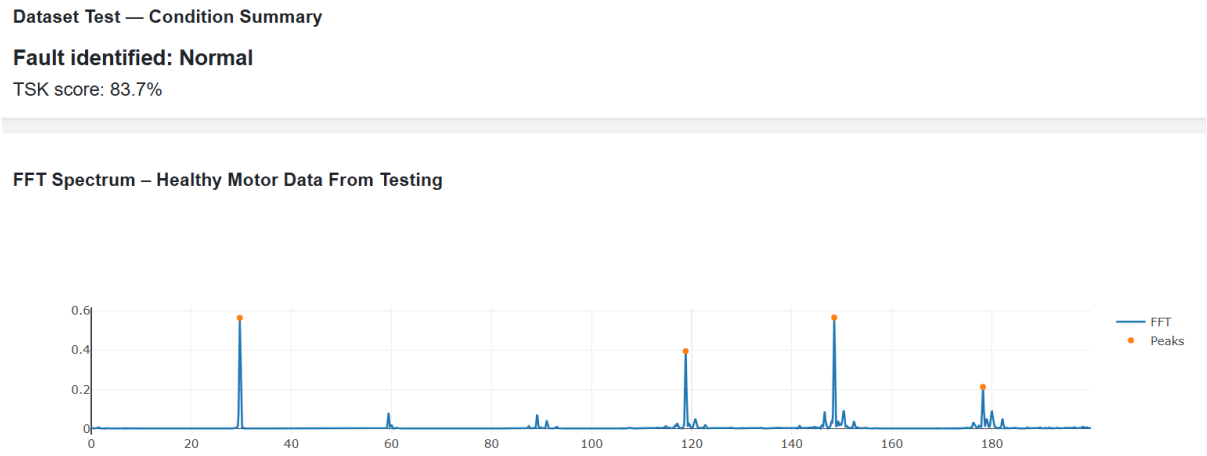


Figure 12: Healthy Motor Data

The lack of faults identified was correct and the motor, post-serviced, was in normal condition. The TSK score was 83.7%. These results showed that the trained model identified the correct fault with a high confidence level.

FFT spectrum demonstrated a prominent peak at approximately 30Hz, which was the fundamental rotational frequency of the motor. Additional peaks were observed at the higher-order harmonics. However, presence of such harmonics may be due to structural characteristics or perhaps, operating conditions.

8.4.2 Frequency-spectrum Output Analysis of Motor with Imbalance

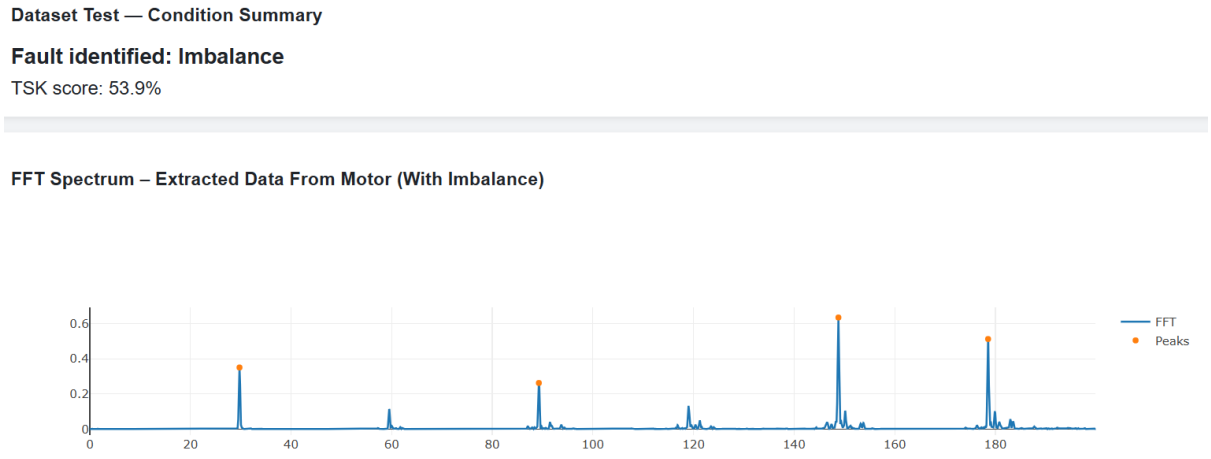


Figure 13: Extracted Imbalance Data

The fault identified was correct, as the data extracted and analysed by AltGuard was from a motor with imbalance. The TSK score is 53.9%. While the correct fault was identified, the relatively low TSK confidence score shows that the imbalance condition had less decisive membership support as compared to that of the healthy motor.

FFT spectrum demonstrated prominent peaks at 30 Hz and multiple harmonics. However, the amplitude at 30Hz was lower than the healthy motor. In addition, the peaks at multiple harmonics were more pronounced, especially the peak at around 90 Hz, the third harmonic. These differences can be explained by the imbalance resulting in distribution of vibration energy across the spectrum.

8.4.3 Significance of Findings

The findings show that AltGuard was able to assign the correct operating conditions, demonstrating its ability in the trained TSK model. Additionally, it was able to generate a fuzzy support score which provides more diagnostic feedback rather than just a normal or abnormal condition. However, more training can be done, to achieve higher confidence level for fault identification.

Furthermore, it demonstrates the feasibility in combining TSK inference with conventional FFT-based feature extraction via frequency-spectrum output, for condition monitoring of rotating machinery.

However, as the data for the motor with imbalance was extracted, the functionality of AltGuard may differ when used on a complete hardware system, with varying operating conditions.

8.5 Discussion

The experimental results provides preliminary support for the feasibility of AltGuard as an AI-assisted condition-monitoring system. However, there were some limitations identified during testing:

Firstly, the limited number of physical test motors. Physical testing was only done on one motor, which restricts the generalisability of the results. Different motors may have varying operating conditions.

Secondly, limited repeated measurements. The benchmark exercise provides paired measurements of at the different locations. However, more readings could be taken at all three locations, before taking an average. This would increase reliability of results.

Thirdly, restrictions on sensor placement. The motor body was circular and there were not many flat surfaces to place AltGuard. Whereas AltGuard's base was flat and required another flat surface on the motor's body to be properly secured.

Fourthly, limited reference instrumentation. The benchmark used was a single-axis vibration meter, that was only able to give figures for vibration velocity. However, a better benchmark would be one that can provide a graph too, and that should be used in future works.

Lastly, limited physical fault testing. The tests were mainly done on the healthy motor. As no testing was done on a faulty motor to see if it can correctly identify the motor's fault, there are is a lack of physical fault testing. This would require the opportunity in testing a faulty motor to evaluate diagnostic abilities, while the motor is running.

Overall, the results provide a good foundation for future testing and further development.

9. Conclusion and Future Work

9.1 Conclusion

The project aims to develop AltGuard, a low-cost, AI-assisted vibration-based condition-monitoring device that can diagnose early faults in marine vessel alternators. Equipped with a hybrid fault detection algorithm that combines threshold-based logic and TSK fuzzy inference, AltGuard aims to detect common faults early in alternators, such as, misalignment, imbalance and bearing fault, mitigating possibility of further machine damage.

The developed prototype integrates a tri-axial accelerometer, ADC, ESP32S3 microcontroller as well as a RPi5 to acquire, send and process vibration signals. Through a Flask-based dashboard, extracted features such as, RMS, kurtosis, dominant frequency and vibration velocity, can be visualised to provide operators with statistics about the alternator. Furthermore, the system also incorporates a hybrid classification system, combining the traditional threshold-based logic with TSK fuzzy inference to provide rapid, early diagnosis that can identify potentially devastating faults before it occurs. This saves significant repair costs for the vessel management and prevents major physical accidents arising from sudden faults. Additionally, the dashboard is able to portray an accurate FFT spectrum, which can identify peaks, supporting the fault identified through TSK inference and associated confidence level.

AltGuard was designed to be reduce water and dust ingress through O-rings between mating surfaces, mainly, the lid and enclosure, as well as, the switch and outer surface of the enclosure.

Through experimental testing, results collected were able to provide preliminary evidence of the prototype's measurement capacity. When benchmarked against a commercial-grade vibration meter, the vibration velocity collected was slight different with the percentage difference ranging from -1.25% to +1.43%. This corresponds to an absolute percentage difference of approximately 0.97% across all six comparable measurements. These results demonstrated a close agreement between the vibration collected by AltGuard and commercial equipment.

During the next phase of testing, the vibration readings from the motor, as well as, that acquired from an online dataset of faulty motor readings, were used to evaluate the condition-monitoring functionality. Both data were processed through AltGuard's FFT analysis and TSK classification model. The system was successful in identifying both healthy and specific faulty conditions, where the post-serviced motor with Normal condition had a TSK confidence score of 83.7%, while the selected dataset from a motor with imbalance was identified as Imbalance with a TSK confidence score of 53.9%.

In conclusion, the project achieved in developing a functional prototype that can acquire vibration data, transmit and process that data, performing frequency-domain analysis and AI-assisted condition assessment via TSK inference model. Although further testing and development is required, AltGuard establishes a foundation for developing an affordable predictive-maintenance solution for marine alternators and other rotating machinery.

9.2 Future Work

9.2.1 Long-term On-site Testing

Future work should prioritise long-term deployment of AltGuard on a vessel or on-land plant, preferably for a period of at least a year. This would allow the system to monitor actual alternators and other rotating machinery under realistic operating conditions. This would provide more data, increasing the generalisability of results from the testing.

Furthermore, additional testing aids in evaluating AltGuard's stability and consistency. This can help in further refining of the hybrid fault-detection classification system and improve its accuracy in predicting faults earlier.

9.2.2 Improvement of TSK Confidence Estimation

Although the TSK fuzzy inference model was trained and consequently, able to identify machine conditions during testing, further refinement can be done to improve its reliability.

Further work should be focused on reviewing the membership functions and rule weights used during the inference process. Membership parameters can also be refined by using the large and diverse collection of data from long-term onsite testing as suggested in 9.2.1. This helps train the model further to provide more meaningful and accurate diagnostic information.

9.2.3 Cybersecurity Improvements

As AltGuard supports transmission of data via wireless communications and a web-based dashboard, cybersecurity is an important valid concern for future development to protect data stored in the Pi, especially when deployed in industrial setting.

Transmission of data through ESP32S3 microcontroller to Pi via a convenient network, introduces potential cybersecurity risks, such as unauthorised access and interception of data. This could prevent successful accurate fault identification and even disrupt communication between system components.

Further development can implement security protocols such as Transport Layer Security (TLS), that encrypts data that is transmitted between the ESP32 and RPi5 (Turner, 2014). Authentication mechanisms can also be added to ensure only authorised individuals can access the system, especially since AltGuard's system is intended to support remote monitoring over external networks.

For industrial applications, be it vessel or on-land plants, cybersecurity is of utmost importance as the system links to the wider vessel or plant communication systems and network. A cybersecurity hack via the communication between ESP32 and AltGuard could infiltrate and extract data within the wider vessel or plant network system. As such, it is

important to ensure safeguards are properly implemented before AltGuard is connected to the vessel or plant networks.

9.2.4 Integration of Thermal Monitoring

To further develop AltGuard and to completement its current diagnostic capabilities based on vibration, thermal monitoring can be integrated with AltGuard.

Temperature is an important indicator of machine condition, especially for rotating components such as bearings (Nembhard et al., 2014). Various factors may contribute to a rise in temperature, such as:

- Mechanical defects
- Excessive friction
- Inadequate lubrication
- Abnormal loading

Monitoring temperature readings alongside vibration characteristics, would provide additional information for evaluating motor health and potentially, improving the accuracy of fault identification.

A temperature sensor, perhaps an external extension attached to the side of the enclosure, can be placed near the bearing housing to measure changes in temperature, and this information is transmitted back to the RPi5 via communication with ESP32S3 microcontroller.

The TSK inference model can also be extended to include temperature as an additional input. A possible scenario such as, an abnormal increase in temperature paired with vibration impulsiveness could provide stronger evidence of bearing-related problems, as compared to using the measurements independently to determine faults.

However, the operating conditions and location of machinery will affect ambient temperature and its load and normal thermal behaviour. Future testing would need to establish thermal baselines before evaluating relationship between thermal variations and the mechanical faults.

The integration would support a more comprehensive, multi-parameter approach to machine condition assessment, that aims to improve the accuracy and reliability of fault identification.

9.2.5 Improvement to Hardware

Further improvements can be done to AltGuard's physical hardware to enhance its durability and ease of use.

An important area for future work would be to send AltGuard for water and dust ingress protection testing in order to determine its IP rating, in accordance to IEC 60529. Consequently, depending on feedback, improvements can be made to ensure an IP rating of at least IP67 is achieved before full-scale industrial deployment.

Another possible future development would be to tackle its current limitation of battery-charging. Currently, the enclosure lid has to be completely removed to access the battery and charge it. However, it brings inconvenience to users. A possible improvement would be to create a slot in the enclosure where the charging port can be placed. However, it would have to be properly secured to prevent water and dust ingress. The charging system can also further incorporate electrical safeguards to prevent overheating, overcharging and short-circuits.

9.3 Potential Real-World Impact

Developing AltGuard has potential applications in marine, industrial and other sectors that rely on rotating machinery for operations.

One of the key impacts is AltGuard's affordability. It remains more accessible as compared to other commercial condition-monitoring systems that are more expensive, limiting its deployment to smaller operators. By utilising comparatively accessible hardware components and a Python-based signal-processing algorithm, AltGuard can provide a cost-effective solution that can be potentially deployed to other rotating machineries, aside from alternators.

In the long run, through the integration of vibration-based and thermal-based monitoring, along with improved AI-assisted fault classification, operators can transit from time-based maintenance to condition-based predictive-maintenance strategies, potentially saving large costs from eventual repair or operating downtime. Reduce time-based maintenance can also save unnecessary maintenance activities, saving time while improving the system's accuracy in early detection of faults.

For now, these potential real-world impacts lie on future testing and developmental works to improve long-term validation and reliable fault identification. With further development and collaboration with industrial partners for on-site testing, AltGuard has the potential to evolve into scalable condition-monitoring system to be used on any rotating machinery. Development of AltGuard reflects how relatively low-cost sensing setup can be integrated with a frequency-domain signal processing and an AI-assisted inference model, to support an accessible, data-drive predictive maintenance.

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11. Appendix

A1) More information on Alternator Repair

The alternator is first lifted out from a tight engine room in a vessel or from a corner of the plant. Both locations require a lot of manpower, and it is dangerous, as one wrong lift could result in loss of lives or physical damage to the alternator. Afterwards, transport arrangements are made to a workshop. Then, the damaged alternator is repaired within a timeframe of usually 2 weeks. Following this, the repaired alternator is sent back to the vessel/plant for reinstatement and commissioning. The whole process is costly and time consuming (RTS Marine, n.d.).

A2) Further information on On-Machine Processing (Edge computing)

Signal preprocessing

This is implemented in processing.py within extract_features() function:

```
signal = signal - np.mean(signal)

window = np.hanning(N)

windowed = signal * window
```

This removes DC offset from the signal and applies a Hanning window to reduce spectral leakage. The Hanning window was chosen instead of other windows as it is able to strike a balance between leakage reduction and frequency resolution, thus making it a strong choice for vibration-based condition monitoring (Harris, 1978).

Feature extraction

Feature extraction is also implemented within preprocessing.py:

```
rms = np.sqrt(np.mean(signal**2))

kurt = kurtosis(signal, fisher=False)

fft_vals = np.fft.fft(windowed, n=N_fft)

freqs = np.fft.fftfreq(N_fft, 1/fs)[:N_fft//2]
```

From these, the system is able to compute RMS, kurtosis, dominant frequency and harmonic ratio.

RMS measures the overall energy level of the vibration signal. The higher the RMS, the more intense the vibration, which is useful for detecting changes in load, which will be covered next, or in detecting severity of vibration.

Kurtosis measures how “peaked” or “flat” the distribution is. In vibration analysis, high kurtosis indicated high impulsiveness of the occurring event, for example, bearing fault.

The bottom two codes are for FFT. “fft_vals” gives the amplitude spectrum, while “freqs” gives the frequency bins. This converts the time-domain signal into frequency domain, revealing dominant vibration frequencies, which can be used to classify faults.

Transient and Steady-state detection

Transient and steady-state detection are implemented in processing.py under detect_transient():

```
cv = np.std(rms_history) / mean_rms
return "Transient" if cv > cv_threshold else "Steady"
```

CV stands for coefficient of variation. This normalises variability relative to the average. This means that if the vibration level fluctuates greatly, this shows the machinery is in an unstable, changing transient state. Whereas, in a steady-state, it is in stable operation, where the CV should be below the threshold.

This determines whether the machine is operating in steady-state or transient conditions.

Spectral analysis was also considered to determine state of machinery, where FFT is used to detect frequency-domain changes, but it is heavier computationally. The CV method is lightweight and easy to implement, while effective for real-time monitoring (Jalilibal, Amiri, Castagliola, & Khoo, 2021).

Fault classification (Threshold-based and fuzzy logic)

Threshold-based classification

This is implemented in processing.py as well and is handled by threshold_classification():

```
if thresholds and thresholds.trained:
    rms_th = thresholds.get_threshold('rms', k=3.0)
    kurt_th = thresholds.get_threshold('kurtosis', k=3.0)
    harm_th = thresholds.get_threshold('harmonic_ratio', k=3.0)
else:
    rms_th = config.FALLBACK_RMS_HIGH
    kurt_th = config.FALLBACK_KURT_HIGH
    harm_th = config.FALLBACK_HARMONIC_HIGH
```

“thresholds” is an instance of AdaptiveThresholds from utilis.py. It learns baseline statistics for mean and standard deviation from a training period when the machine is healthy. When “thresholds” exist and finished training, the dynamic thresholds can be computed and here k=3 means the alarm level is 3 standard deviations above healthy mean, which is a common

statistical process control limit. However, if no adaptive thresholds are available, the code falls back on hard-coded constants from config.py, as a safety net.

Fixed thresholds might not work for all machines as such, adaptive thresholds are in place to allow the system to calibrate itself to the true vibration levels of that specific machine.

```
    confidences = {}

    if features['kurtosis'] > kurt_th:

        confidences['Bearing Fault'] = min((features['kurtosis'] - kurt_th) /
kurt_th, 1.0)

    if features['harmonic_ratio'] > harm_th:

        confidences['Misalignment'] = min((features['harmonic_ratio'] -
harm_th) / harm_th, 1.0)

    if features['rms'] > rms_th and features['harmonic_ratio'] < 0.2:

        confidences['Imbalance'] = min((features['rms'] - rms_th) / rms_th,
1.0)
```

For each fault type, we check if the relevant feature exceed thresholds which computes a confidence between 0 and 1.0, showing how far the value exceeds the thresholds.

For ‘imbalance’, there is an extra condition harmonic_ratio < 0.2. This is because a high RMS alone could be caused by misalignment or bearing fault. Misalignment presents itself with a strong 2x harmonic, but if the harmonic ratio is low, such as in the above condition, then imbalance is likely the cause instead.

TSK Fuzzy Inference System Classification

The fuzzy logic is implemented in fuzzy.py. Some of the vital codes will be displayed and explained below.

```
def gaussian(x, mean, sigma):

    """Gaussian membership function."""

    return np.exp(-0.5 * ((x - mean) / sigma) ** 2)
```

The classic Gaussian bell curve is used for AltGuard as it is smooth, continuous and also has natural decay, which has advantages in representing vague concepts such as “low” or “high” without sharp transitions.

```
    # Rule 1: Low everything → Normal

    w1 = rms_low(rms) * kurt_low(kurt) * harm_low(harmonic)

    rules.append((w1, 0))
```

```
    # Rule 2: High RMS, low harmonic → Imbalance
    w2 = rms_high(rms) * harm_low(harmonic)
    rules.append((w2, 1))
    # Rule 3: High harmonic ratio → Misalignment
    w3 = harm_high(harmonic)
    rules.append((w3, 2))

    # Rule 4: High kurtosis → Bearing fault
    w4 = kurt_high(kurt)
    rules.append((w4, 3))
```

The rule strengths w1 to w4, is the product of membership values.

A zero-order TSK is used as the consequent is a constant and is simpler to tune. Additionally, it is effective for classification.

```
    numerator = sum(w * output for w, output in rules)
    denominator = sum(w for w, _ in rules)
    fault_index = numerator / denominator
```

This gives the weighted average of the rule outputs and is a cript number between 0 and 3.

```
    confidence = denominator / len(rules)
    states = ["Normal", "Imbalance", "Misalignment", "Bearing Fault"]
    final_state = states[int(round(fault_index))]
    return final_state, confidence
```

“confidence” gives us the fraction of total possible rule activation where it measures how well the input matches the rule base.

Finally, the final_state can be deduced from the fault_index, which shows one of the states above, “normal”, “imbalance”, “misalignment” or “bearing fault”.

This approach is very useful for AltGuard as it helps to handle uncertainty and unfamiliar machinery. Additionally, it is a lightweight code, with easy computations that can run easily on the RPi5. Furthermore, the confidence score allows the system to know when it is not certain, hence overriding TSK system with threshold rules during transient states.

A3) Details on Off-Machine Processing

Data Transmission via Flask API

The data generated on the Pi is exposed through a web sever implemented in dashboard.py using Flask and these key API endpoints provide latest classification results and FFT spectrum:

```
@app.route('/api/latest')
```

```
@app.route('/api/history')
```

```
@app.route('/api/fft')
```

Real-Time dashboard visualisation

The dashboard is able to retrieve data using:

```
fetch(‘/api/latest’)
```

```
fetch(‘/api/fft’)
```

These enable the dashboard to display:

- Real-time feature values (RMS, kurtosis, frequency)
- Machine state classification and confidence
- Live FFT spectrum with peak detection
- Time-series trend of vibration features

Asychronous Data Streaming

Real-time updates are handled using Server-Sent Events (SSE):

```
@app.route(‘/api/stream’)
```

This enables continuous data streaming to the dashboard, providing real-time data.

A4) Simulation Results

Graph of FFT Spectrum

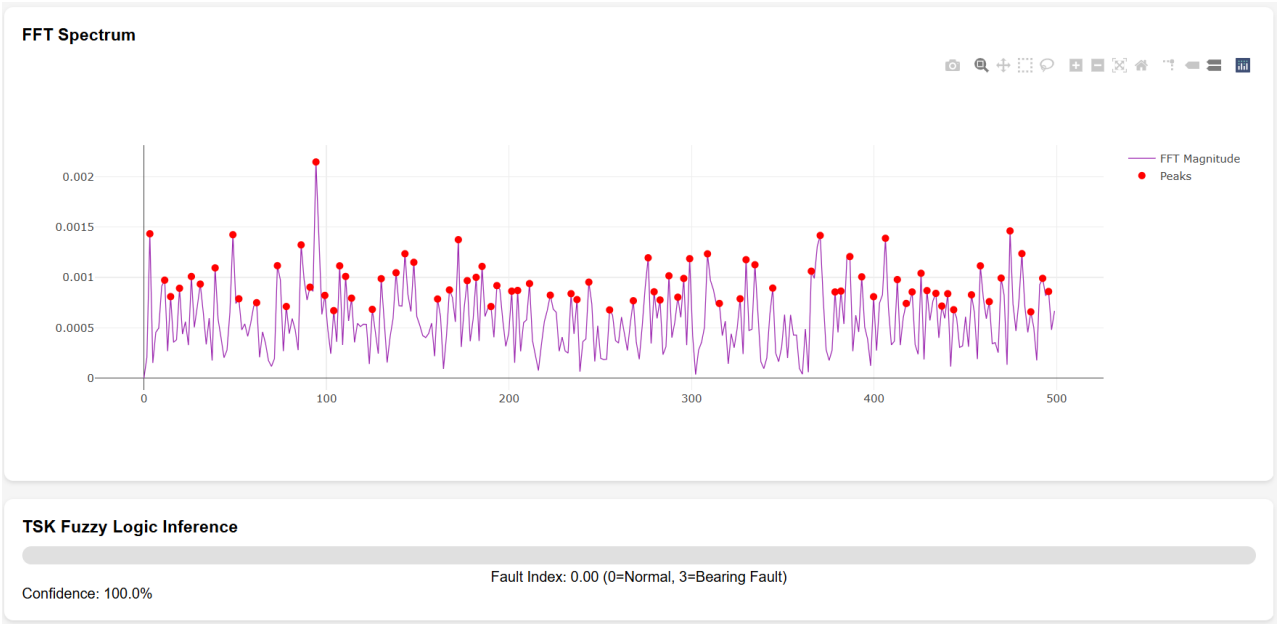


Figure 14: FFT spectrum for simulated “Normal” state

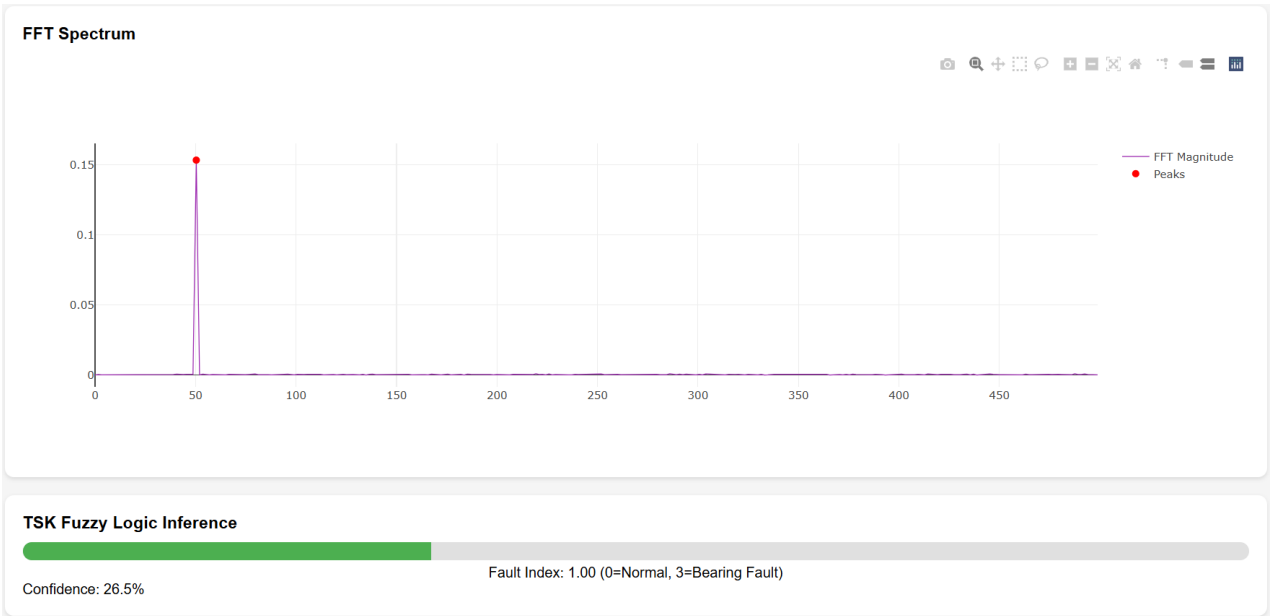


Figure 15: FFT spectrum for simulated “Imbalance” state

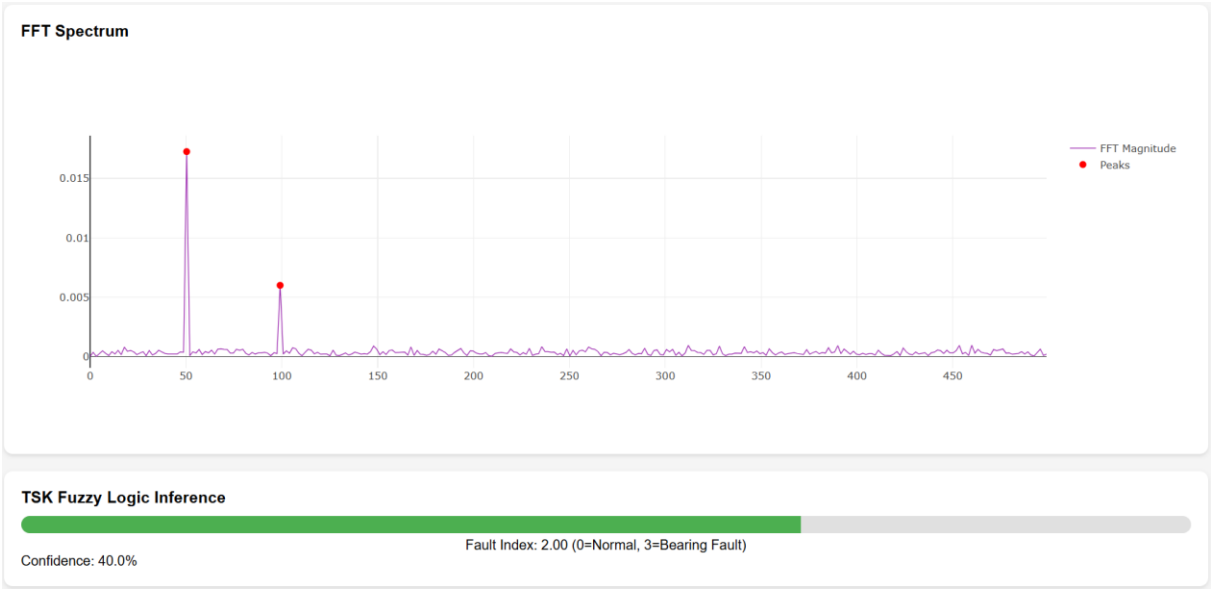


Figure 16: FFT spectrum for simulated “Misalignment” state

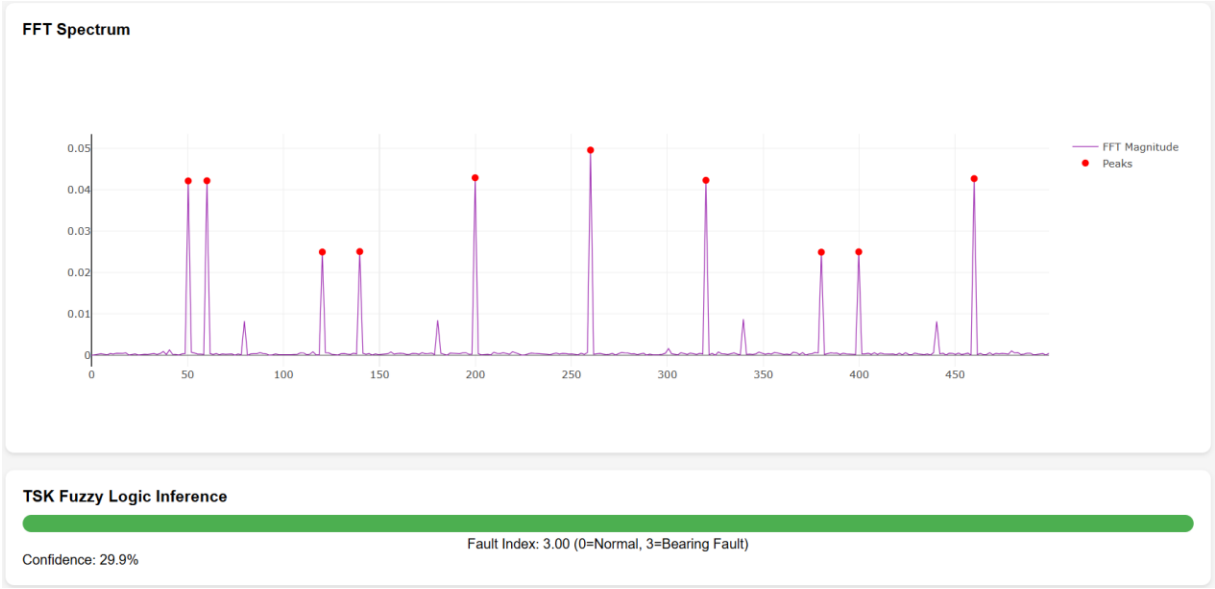


Figure 17: FFT spectrum for simulated “Bearing fault” state

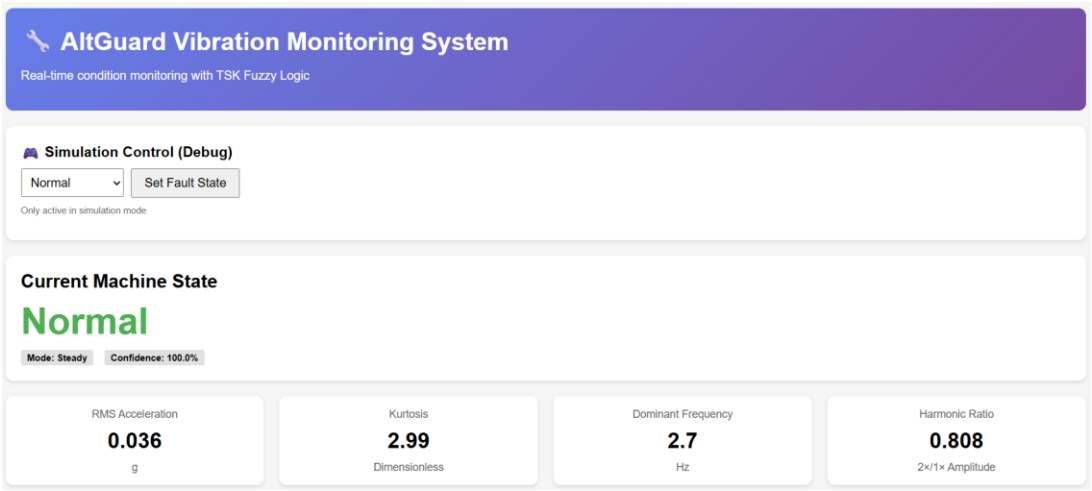


Figure 18: Machine state and feature values for “Normal”

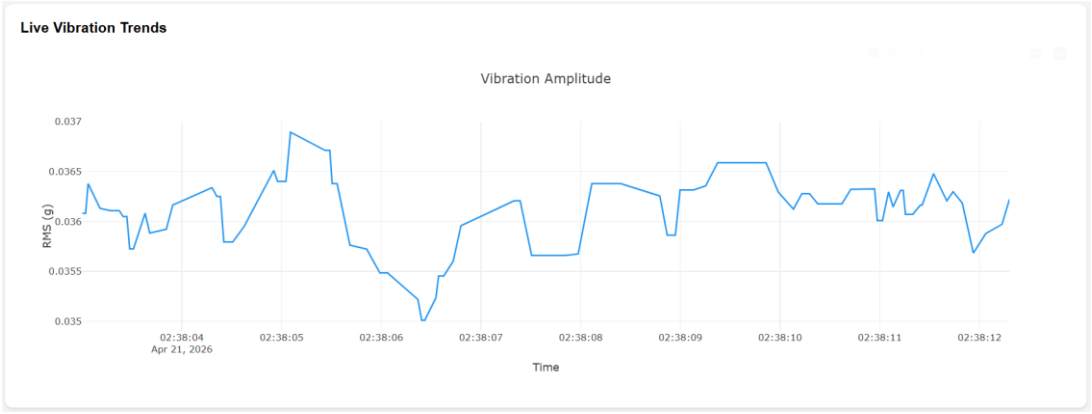


Figure 19: Live vibration trends for “Normal”

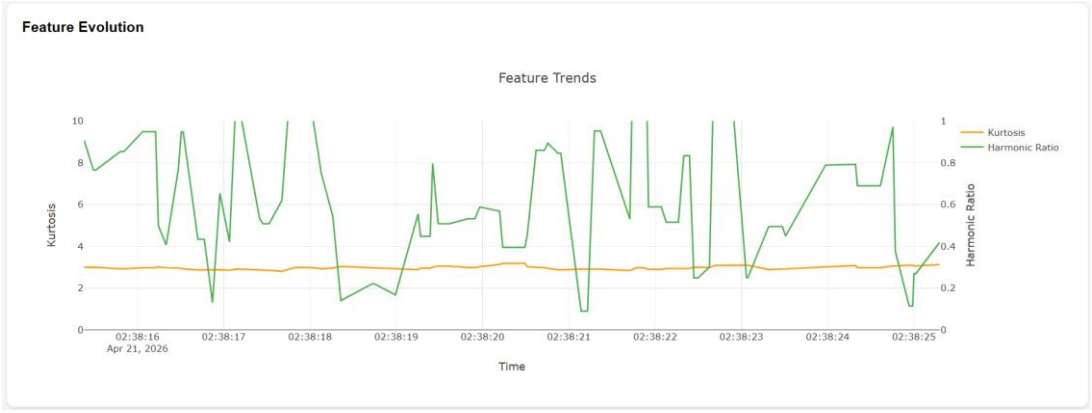


Figure 20: Feature trends for “Normal”

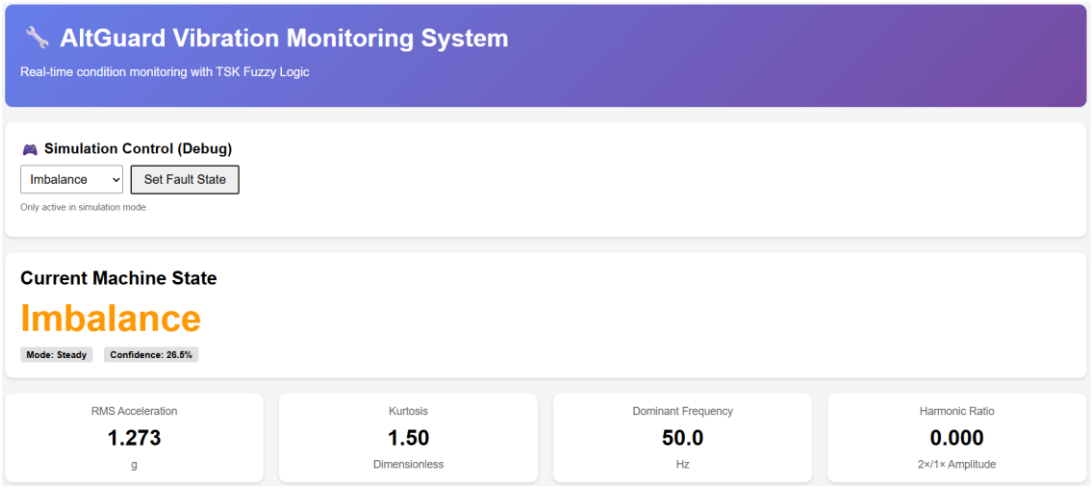


Figure 21: Machine state and feature values for “Imbalance”

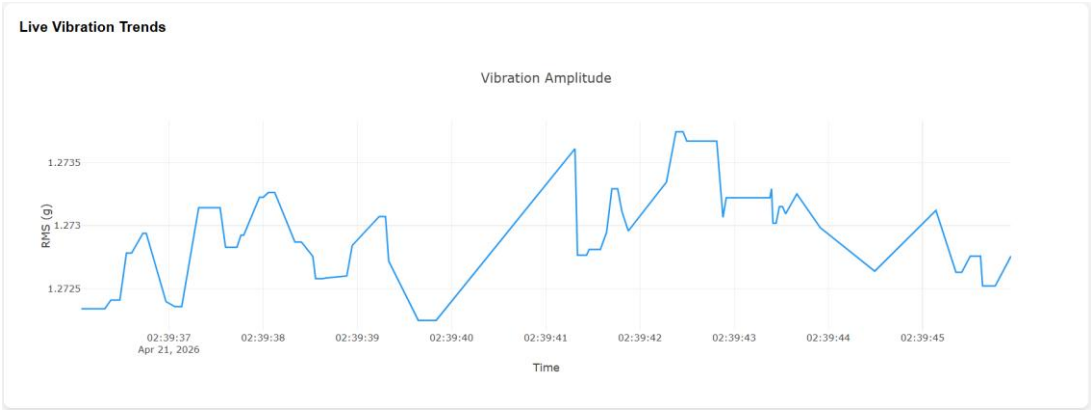


Figure 22: Live vibration trends for “Imbalance”

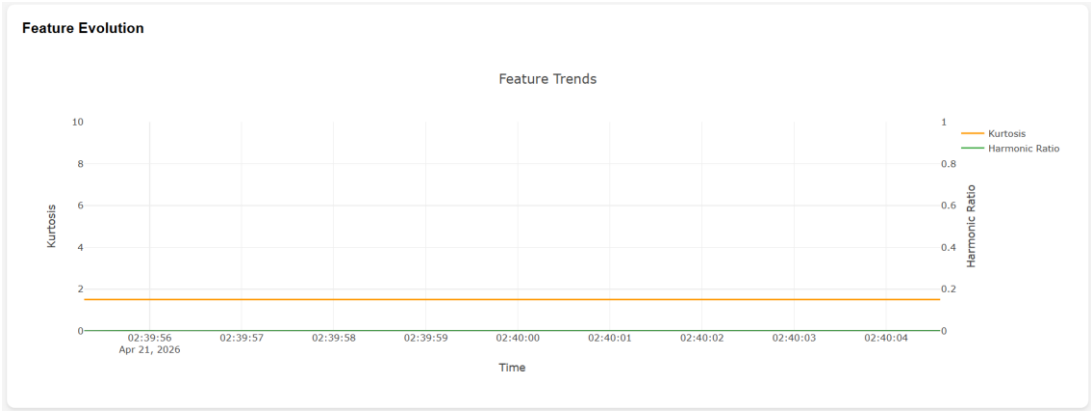


Figure 23: Feature trends for “Imbalance”

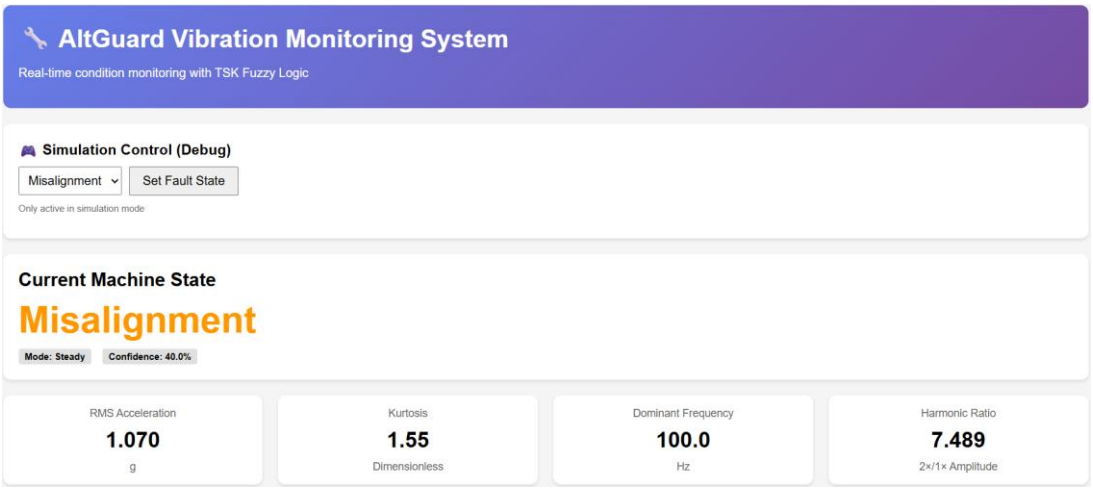


Figure 24: Machine state and feature values for “Misalignment”

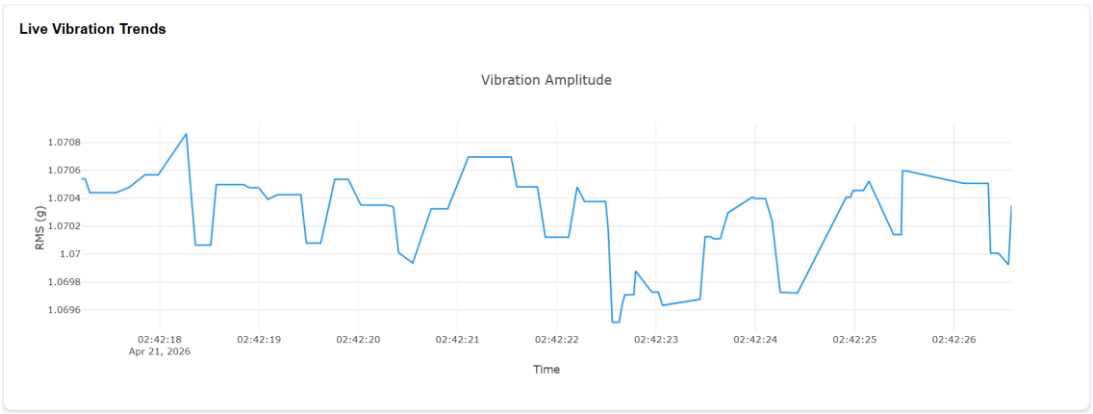


Figure 25: Live vibration trends for “Misalignment”

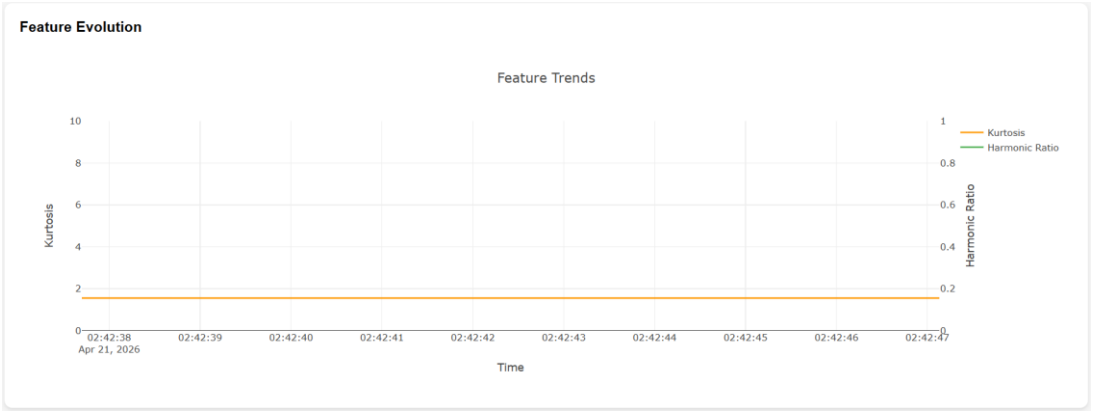


Figure 26: Feature trends for “Misalignment”

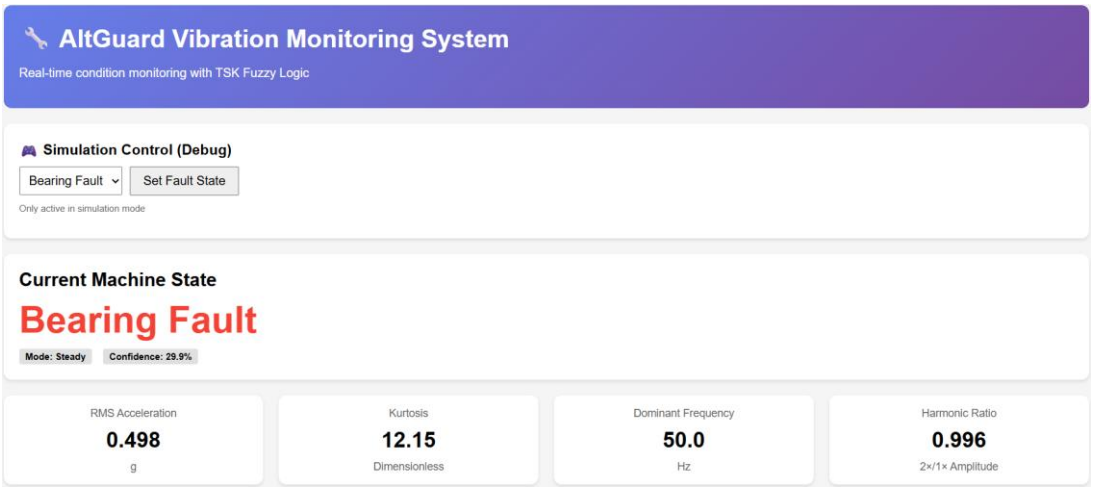


Figure 27: Machine state and feature values for “Bearing Fault”

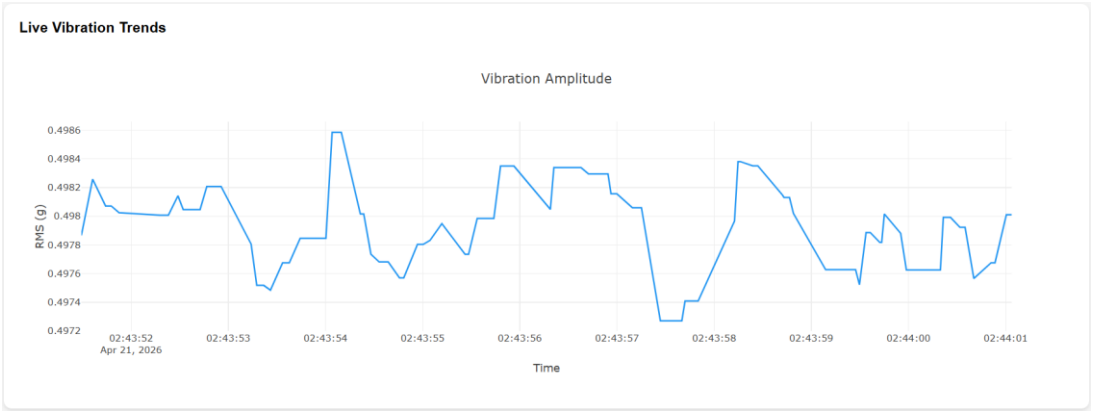


Figure 28: Live vibration trends for “Bearing Fault”

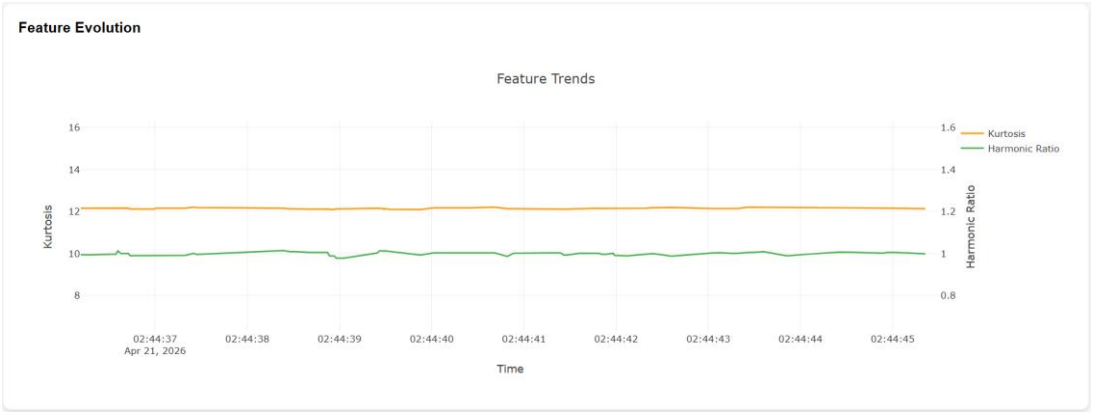


Figure 29: Feature trends for “Bearing Fault”

From these FFT spectrum graphs, the correct peaks associated with the relevant faults were identified. However, the confidence levels for the TSK inference system is not as high as expected. This limitation is because the simulations were based off predefined functions, that only simulates vibration signals, approximating real-world vibration faults signatures. In

order to improve confidence levels, further improvements are needed to the TSK system, which will be done in the subsequent half of the FYP.